



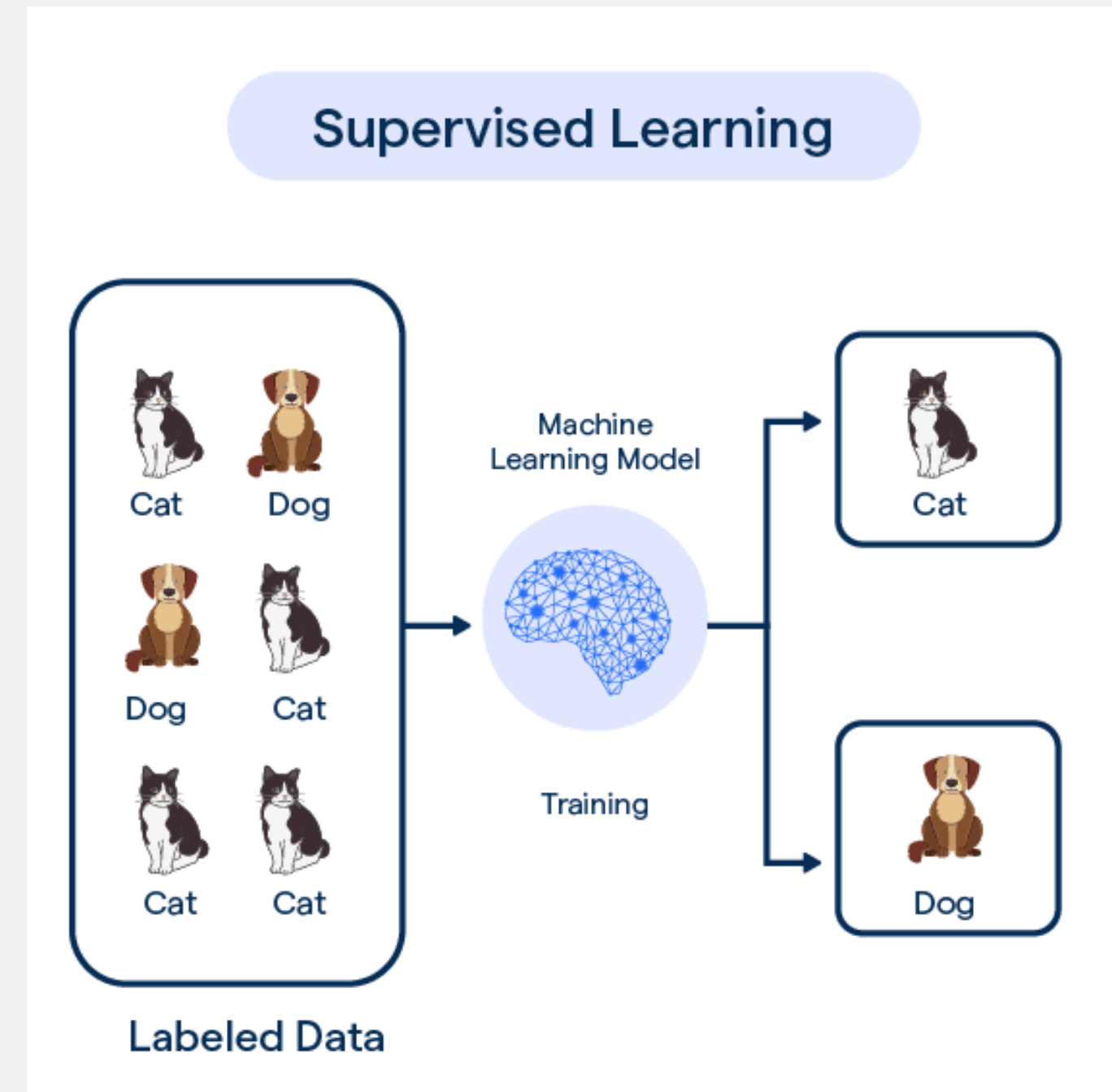
# Supervised Learning

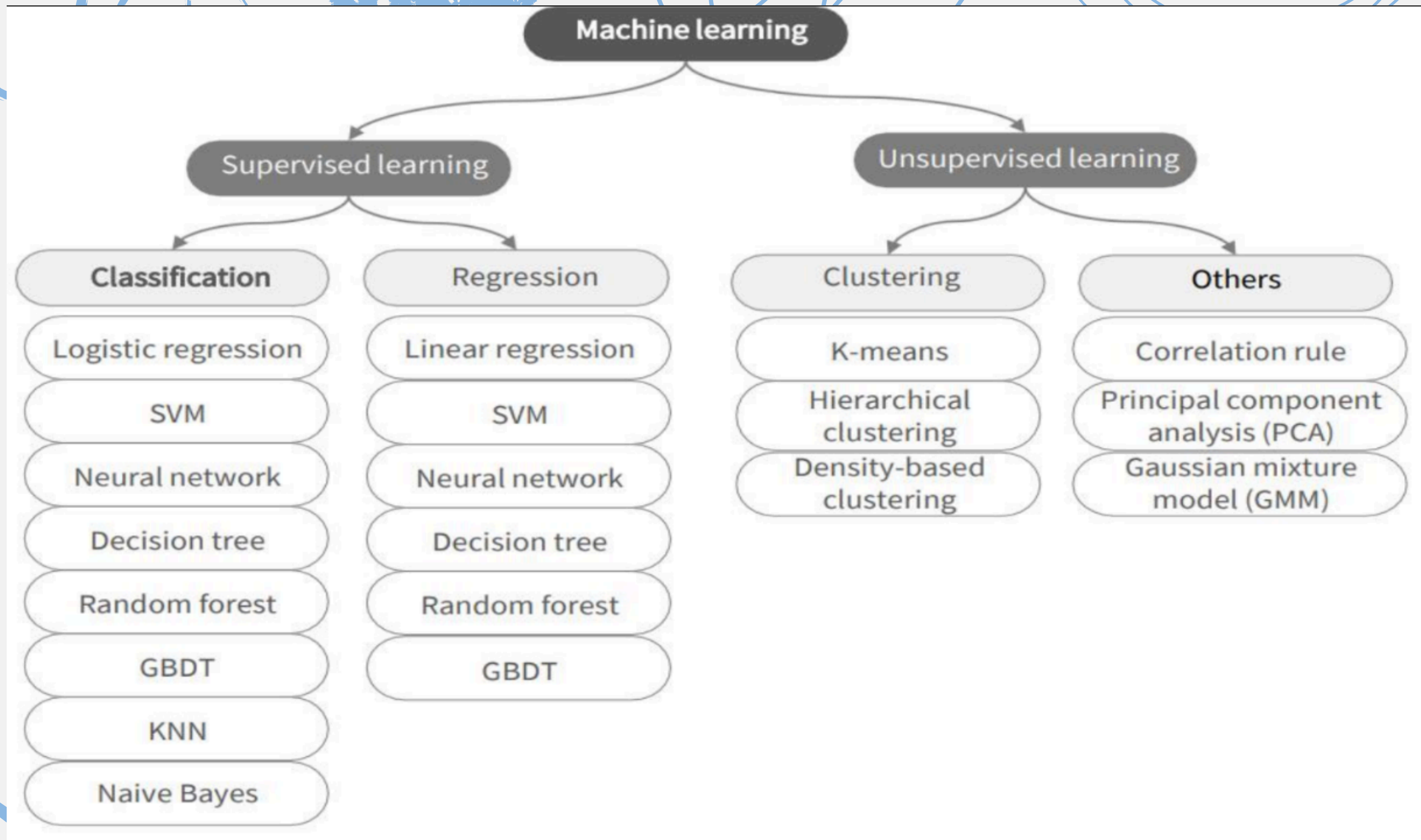
**MSP'26 ML Module Session02**



# Supervised Learning

Supervised learning is a type of machine learning where a model learns from labelled data—meaning every input has a corresponding correct output. The model makes predictions and compares them with the true outputs, adjusting itself to reduce errors and improve accuracy over time. The goal is to make accurate predictions on new, unseen data.

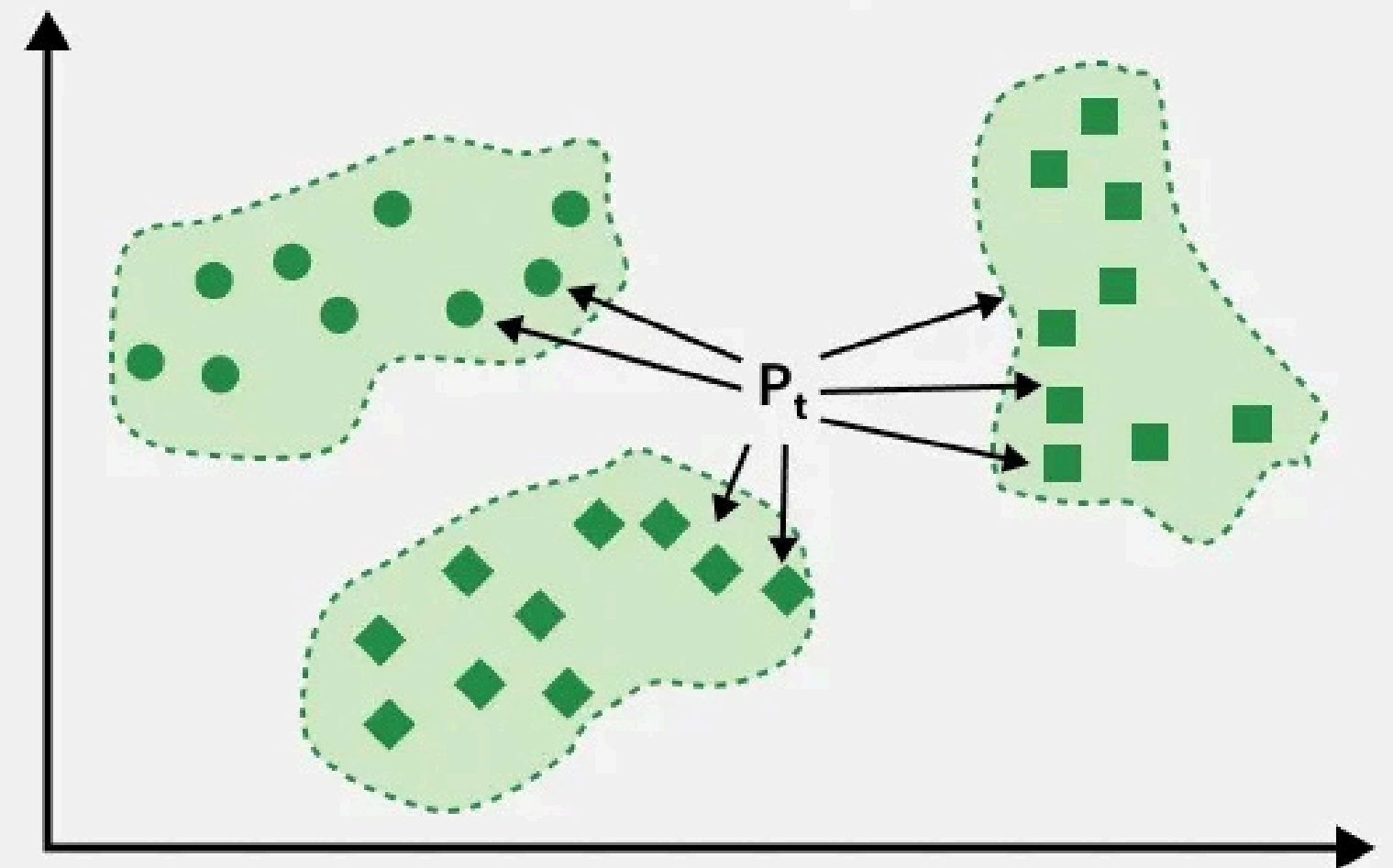




# K-nearest neighbours (knn)

1. Initialize  $k$  to your chosen number of neighbours
2. For each sample in the data:
  - Calculate the distance between the query example and the current example from the data
  - Add the distance and the index of the example to an ordered collection
3. Sort the ordered collection of distances and indices from smallest to largest

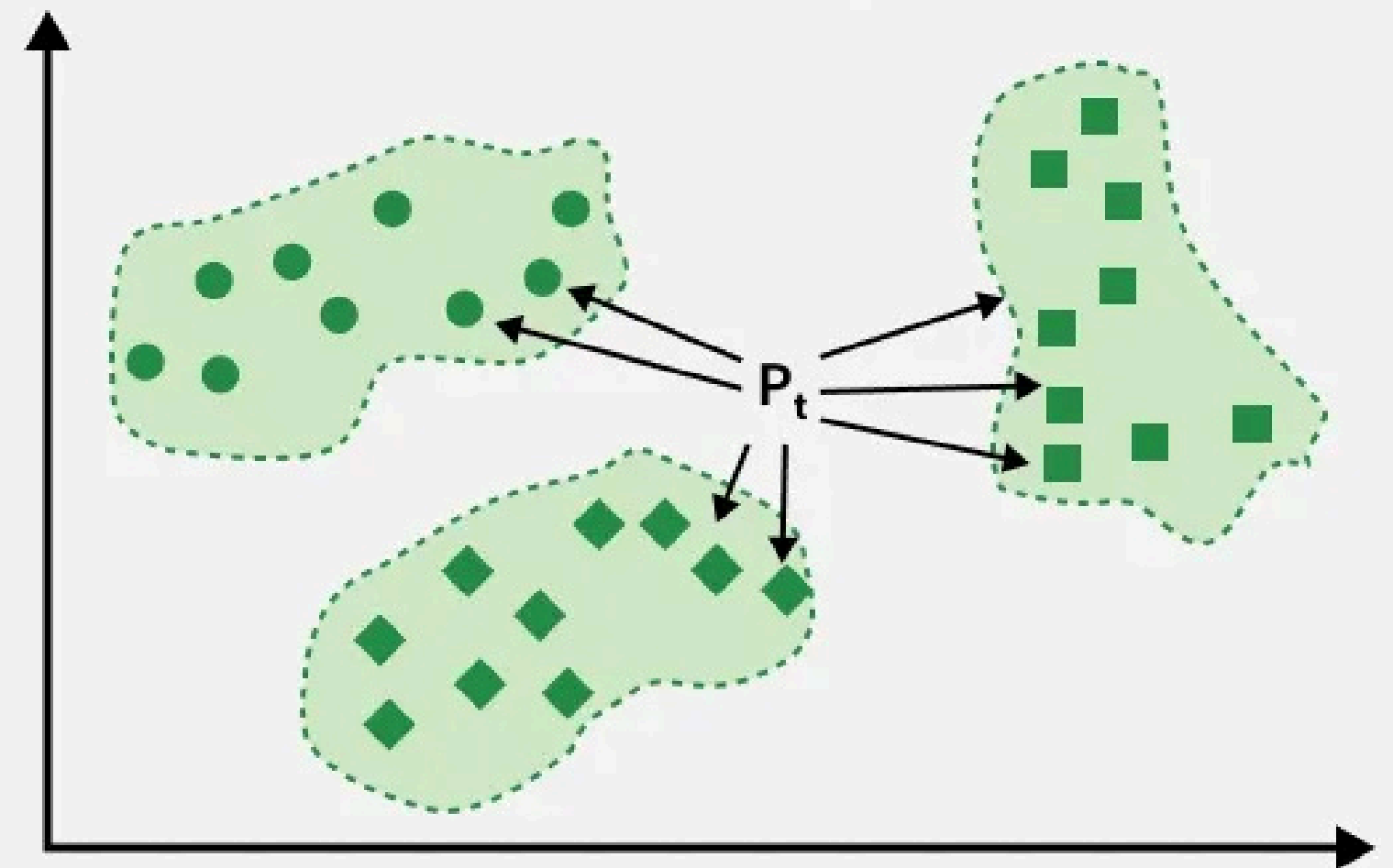
## K Nearest Neighbors



# K-nearest neighbours (knn) cont'd

4. Sort the ordered collection of distances and indices from smallest to largest (in ascending order) by the distances
5. Pick the first K entries from the sorted collection
6. Get the labels of the selected K entries
7. if regression, return the mean of the K labels
8. If classification, return the mode of the K labels (if using majority voting)

## K Nearest Neighbors





# knn – classification

| Training Point (X1, X2) | Class | Distance from (3,7) | Calculation                                         | Rank |
|-------------------------|-------|---------------------|-----------------------------------------------------|------|
| (7, 7)                  | Bad   | 4.00                | $\sqrt{[(7-3)^2 + (7-7)^2]} = \sqrt{(16+0)} = 4.00$ | 3    |
| (7, 4)                  | Bad   | 5.00                | $\sqrt{[(7-3)^2 + (4-7)^2]} = \sqrt{(16+9)} = 5.00$ | 4    |
| (3, 4)                  | Good  | 3.00                | $\sqrt{[(3-3)^2 + (4-7)^2]} = \sqrt{(0+9)} = 3.00$  | 1    |
| (1, 4)                  | Good  | 3.61                | $\sqrt{[(1-3)^2 + (4-7)^2]} = \sqrt{(4+9)} = 3.61$  | 2    |



# knn – classification cont'd

## Step 3: Select $K = 3$ nearest neighbors

| Neighbor           | Class |
|--------------------|-------|
| 1st nearest: (3,4) | Good  |
| 2nd nearest: (1,4) | Good  |
| 3rd nearest: (7,7) | Bad   |

## Step 4: Majority voting

| Class | Count |
|-------|-------|
| Good  | 2     |
| Bad   | 1     |

# knn - regression

| ID | Height | Age | Weight | Distance from<br>(5.5,38) | Calculation                                           | Rank |
|----|--------|-----|--------|---------------------------|-------------------------------------------------------|------|
| 1  | 5.0    | 45  | 77     | 7.02                      | $\sqrt{[(5.0-5.5)^2+(45-38)^2]} = \sqrt{(0.25+49)}$   | 9    |
| 2  | 5.11   | 26  | 47     | 12.00                     | $\sqrt{[(5.11-5.5)^2+(26-38)^2]} = \sqrt{(0.15+144)}$ | 10   |
| 3  | 5.6    | 30  | 55     | 8.00                      | $\sqrt{[(5.6-5.5)^2+(30-38)^2]} = \sqrt{(0.01+64)}$   | 7    |
| 4  | 5.5    | 34  | 59     | 4.00                      | $\sqrt{[(5.5-5.5)^2+(34-38)^2]} = \sqrt{(0+16)}$      | 2    |
| 5  | 4.8    | 40  | 72     | 2.12                      | $\sqrt{[(4.8-5.5)^2+(40-38)^2]} = \sqrt{(0.49+4)}$    | 1    |
| 6  | 5.8    | 36  | 60     | 2.02                      | $\sqrt{[(5.8-5.5)^2+(36-38)^2]} = \sqrt{(0.09+4)}$    | 0    |
| 7  | 5.3    | 19  | 40     | 19.00                     | $\sqrt{[(5.3-5.5)^2+(19-38)^2]} = \sqrt{(0.04+361)}$  | 11   |
| 8  | 5.8    | 28  | 60     | 10.00                     | $\sqrt{[(5.8-5.5)^2+(28-38)^2]} = \sqrt{(0.09+100)}$  | 8    |
| 9  | 5.5    | 23  | 45     | 15.00                     | $\sqrt{[(5.5-5.5)^2+(23-38)^2]} = \sqrt{(0+225)}$     | 12   |
| 10 | 5.6    | 32  | 58     | 6.00                      | $\sqrt{[(5.6-5.5)^2+(32-38)^2]} = \sqrt{(0.01+36)}$   | 6    |



# knn - regression cont'd

Nearest 3 distances:

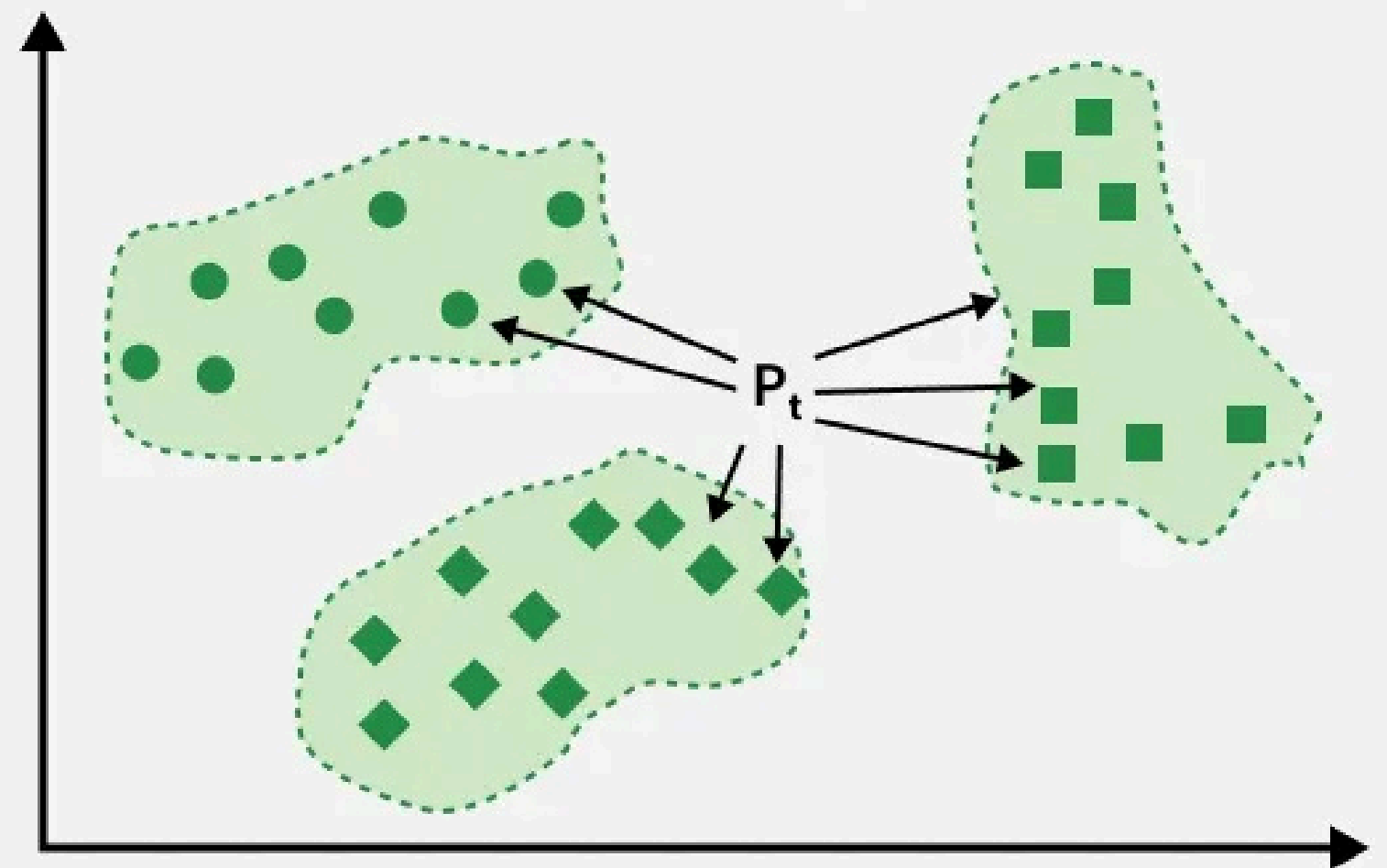
| ID | Weight | Distance |
|----|--------|----------|
| 6  | 60     | 2.02     |
| 5  | 72     | 2.12     |
| 4  | 59     | 4.00     |

$$\begin{aligned} \text{Predicted Weight} &= \frac{60 + 72 + 59}{3} \\ &= \frac{191}{3} \\ &= 63.67 \end{aligned}$$

# knn – how to pick optimal k

- **Small K values:** This can make the model highly sensitive to noise and might overfit the training data
- **Large K values:** This can make the model smooth out the decision boundary, leading to underfitting. The classifier becomes less sensitive to local patterns and more influenced by the overall data distribution.

## K Nearest Neighbors



Search Task: how to pick perfect k?

# Naive Bayes Classifier

## LIKELIHOOD

The probability of "B" being True, given "A" is True

## PRIOR

The probability "A" being True. This is the knowledge.

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)}$$

## POSTERIOR

The probability of "A" being True, given "B" is True

## MARGINALIZATION

The probability "B" being True.

# Naive Bayes – classification

| Example No. | Color  | Type   | Origin   | Stolen? |
|-------------|--------|--------|----------|---------|
| 1           | Red    | Sports | Domestic | Yes     |
| 2           | Red    | Sports | Domestic | No      |
| 3           | Red    | Sports | Domestic | Yes     |
| 4           | Yellow | Sports | Domestic | No      |
| 5           | Yellow | Sports | Imported | Yes     |
| 6           | Yellow | SUV    | Imported | No      |
| 7           | Yellow | SUV    | Imported | Yes     |
| 8           | Yellow | SUV    | Domestic | No      |
| 9           | Red    | SUV    | Imported | No      |
| 10          | Red    | Sports | Imported | Yes     |

*New Instance = (Red, SUV, Domestic)*



# Naive Bayes – classification cont'd

$$p(Yes) = \frac{5}{10} = 0.5$$

$$p(No) = \frac{5}{10} = 0.5$$

| Color  | Yes | No  |
|--------|-----|-----|
| Red    | 3/5 | 2/5 |
| Yellow | 2/5 | 3/5 |

| Type   | Yes | No  |
|--------|-----|-----|
| Sports | 4/5 | 2/5 |
| SUV    | 1/5 | 3/5 |

| Origin   | Yes | No  |
|----------|-----|-----|
| Domestic | 2/5 | 3/5 |
| Imported | 3/5 | 2/5 |

$$P(Yes|New Instance) = p(Yes) * P(Color = Red|Yes) * P(Type = SUV|Yes) * P(Origin = Domestic|Yes)$$

$$P(Yes|New Instance) = \frac{5}{10} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} = \frac{3}{125} = 0.024$$

$$P(No|New Instance) = p(No) * P(Color = Red|No) * P(Type = SUV|No) * P(Origin = Domestic|No)$$

$$P(No|New Instance) = \frac{5}{10} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{9}{125} = 0.072$$

# Naive Bayes – regression

| Class   | Temperature |
|---------|-------------|
| Healthy | 36.5        |
| Healthy | 36.7        |
| Healthy | 36.8        |
| Sick    | 38.0        |
| Sick    | 38.2        |
| Sick    | 38.4        |



# Naive Bayes – regression cont'd

**Prior**

**x**

**Likelihood**

$$P(C_1) = \frac{3}{6} = 0.5$$
$$P(C_2) = \frac{3}{6} = 0.5$$

$$P(x|C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right)$$

# Naive Bayes – regression cont'd

Healthy ( $C_1$ )

$$\mu_1 = \frac{36.5 + 36.7 + 36.8}{3} = 36.67$$

$$\sigma_1^2 = \frac{(36.5 - 36.67)^2 + (36.7 - 36.67)^2 + (36.8 - 36.67)^2}{3}$$

$$\sigma_1^2 \approx 0.0156$$

Sick ( $C_2$ )

$$\mu_2 = \frac{38.0 + 38.2 + 38.4}{3} = 38.2$$

$$\sigma_2^2 = \frac{(38.0 - 38.2)^2 + (38.2 - 38.2)^2 + (38.4 - 38.2)^2}{3}$$

$$\sigma_2^2 \approx 0.0267$$

# Naive Bayes – regression cont'd

Classify 37.5:

Likelihood for Healthy

$$P(x|C_1) = \frac{1}{\sqrt{2\pi(0.0156)}} \exp\left(-\frac{(37.5 - 36.67)^2}{2(0.0156)}\right)$$

$$P(x|C_1) \approx 0.00003$$

Likelihood for Sick

$$P(x|C_2) = \frac{1}{\sqrt{2\pi(0.0267)}} \exp\left(-\frac{(37.5 - 38.2)^2}{2(0.0267)}\right)$$

$$P(x|C_2) \approx 0.11$$

$$P(C_1|x) \propto P(x|C_1)P(C_1) = 0.00003 \times 0.5$$

$$P(C_2|x) \propto P(x|C_2)P(C_2) = 0.11 \times 0.5$$